DEVELOPER SESSION

motionkit

Make motion-graphics videos in Blender 5 with a few lines of Python, or just ask Gemini to write them

[Presenter] · [Team] · [Date]



The trailer taught us what's hard

1,661
lines of Blender Python in the ERM trailer script

25
lines for a polished title shot with motionkit

0
Blender knowledge needed to start

WHAT USED TO BE HARD, NOW HANDLED BY THE KIT

Blender version quirks
Keyframes, compositor, EEVEE and the sequencer changed between 4.x and 5.x: one compatibility file absorbs it

Materials and glow
Self-lit materials with fade, glow and colour you can animate, plus bloom

Camera maths
Say where to stand and what to look at; orbits, push-ins and cuts are one line

Frames and easing
Think in seconds and named eases: "smooth", "back", "bounce"

Rendering
One command for a still, a preview or the final MP4 with music

From an idea to an MP4 in four hops

1
Idea
"A 10-second intro: our name types itself among stars"

→

2
Shot Script
Plain-language plan: what, when, camera, transitions

→

3
Gemini gem
motionkit Director writes the Python for you

→

4
motionkit
Your script: assets, motions, camera, overlay

→

5
Blender 5
EEVEE renders it: still, preview or final MP4

Two ways in. Write the motionkit script yourself (it reads like a storyboard), or describe the video to the gem and paste its code. Either way, the result is a normal Python file you can read, change and keep in git.

Reusable components for everything we needed

Movie
Clean scene, EEVEE settings, shots, music, still / preview / render from the command line

```
mk.Movie("intro", length=10)
```

Assets
Text3D · Counter · Card · Screen · Logo · Sphere · Ring · Arc · Line · Starfield · GridFloor · Globe · BarChart · ScreenWall · Portal · Group

```
mk.Globe().focus(43.6, 1.4, at=2)
```

Motions (every asset)
fade · pop · slide · rise · move · scale · rotate · spin · float · pulse · glow · colour · draw-on · type-on

```
.pop_in(at=0.5).float(1, 6)
```

Camera
place · move · cut · push-in · pull-out · orbit · crane · path · shake · zoom · dolly-zoom · focus

```
camera.orbit(radius=9, at=2)
```

Overlay
caption · title · lower third · text · image · box · letterbox · fades · flash

```
overlay.caption("Hi", at=1)
```

Transitions + effects
cut · fade through black · dip · flash cut · whip pan · push through · through portal · glow · vignette · motion blur

```
tr.flash_cut(movie, at=5, ...)
```

25 lines, 8 seconds of video

```
movie = mk.Movie("hello_title", length=8)

mk.Starfield()
mk.GridFloor(height=-1.6)

ring = mk.Ring(radius=2.6, color="indigo", glow=3).at(0, 1.5, 0.3)
ring.draw_on(at=0.3, duration=1.2).spin(turns=0.25, start=1.5, end=8, axis="Y")

title = mk.Text3D("Hello DMS", size=1.1, color="pink", glow=3, depth=0.12).at(0,
0, 0.3)
title.pop_in(at=0.8, duration=0.7).float(start=1.5, end=8, height=0.08)

movie.camera.place((0, -14, 1.5), look_at=(0, 0, 0.3))
movie.camera.orbit(center=(0, 0, 0.3), radius=11, start=-25, end=12,
                   height=1.0, at=0, duration=8)

movie.overlay.caption("Built with motionkit in 25 lines", eyebrow="Demo", at=3,
until=6.6)
movie.overlay.fade_from_black(at=0, duration=1)
movie.overlay.fade_to_black(at=7.2, duration=0.8)
movie.effects.glow().vignette(0.3)

movie.run()
```



Create · place · animate

Every line is one idea. Times are seconds. Motions chain. The camera looks at a point.

Three commands cover everything

```
blender -b -P examples/01_hello_title.py -- --still 3.5
```

One PNG at 3.5 s. Seconds, so check every change this way

```
blender -b -P examples/01_hello_title.py -- --preview --render
```

Half-size MP4: watch the timing

```
blender -b -P examples/01_hello_title.py -- --render
```

Final 1080p MP4 (with music if added)

```
blender -P examples/01_hello_title.py
```

Open Blender with the scene built: scrub, play, tweak by hand

`-b` no window **`-P`** run the script **`--`** options for motionkit **`--percent 25`** tiny, fast test renders **Output:** `examples/render/`

Camera moves and transitions, one line each

movie.camera.…

`place / move`
stand here, look there; travel

`push_in / pull_out`
closer for emphasis, away to reveal

`orbit`
circle the subject

`crane`
rise or sink

`follow_path`
fly through points

`shake`
impact, handheld

`zoom / dolly_zoom`
lens change, vertigo

`cut_to`
hard cut

tr.…(movie, at=…, to=shot.point(…), look_at=…)

`fade_through_black`
calm chapter change

`dip_to_color`
soft reveal

`flash_cut`
energy, a beat in the music

`whip_pan`
fast sideways smear

`push_through`
rush into the next shot

`through_portal`
fly through a gate (the trailer)

Describe the video; the gem writes the code

YOU TYPE (SHOT SCRIPT)

```
/script
MOVIE "kpi_story" 12s
SHOT data 0s-12s
  ADD chart "bars" 345,120,95,60
      labels B,O,P,A at 0,2,-1.4
  bars: grow @0.5 | highlight 0 @2.5
  ADD counter "users" to 892
      suffix " users" @1 for 1.5s
  CAMERA place 0,-12,1 look 0,1,0
  CAMERA push-in to 0,-9,0.6 @0 for 12s
  CAPTION "Used by 892 people" @3 until 10
EFFECTS glow
```



THE GEM ANSWERS (PYTHON)

```
movie = mk.Movie("kpi_story", length=12)
with movie.shot("data", 0, 12):
    bars = mk.BarChart([345, 120, 95, 60],
        labels=["B", "O", "P", "A"]).at(0, 2, -1.4)
    bars.grow(at=0.5).highlight(0, at=2.5)
    mk.Counter(892, at=1, duration=1.5,
        suffix=" users").at(3.5, 1.5, 0.5)
    movie.camera.place((0, -12, 1), look_at=(0, 1, 0))
    movie.camera.push_in(to=(0, -9, 0.6), at=0, duration=12)
    movie.overlay.caption("Used by 892 people", at=3,
until=10)
movie.effects.glow()
movie.run()
```

/idea	/script	/storyboard	/fix	/tweak	/vertical
free text → Shot Script	Shot Script → Python	shot table	paste an error	"slower", "gold title"	phone format

SET UP THE GEM

Airbus Amber

motionkit Director in five minutes

CREATE

1. **Open Gemini**
   gemini.google.com › Explore Gems (Gem manager) › New Gem
2. **Name and description**
   motionkit Director · copy the description from gemini-gem/GEM.md
3. **Instructions**
   Paste the block from GEM.md ("You are motionkit Director…" to /vertical)
4. **Knowledge**
   Add files › knowledge/motionkit-knowledge.md (rename to .txt if refused)
5. **Model**
   Pick the Thinking model if offered: more accurate code

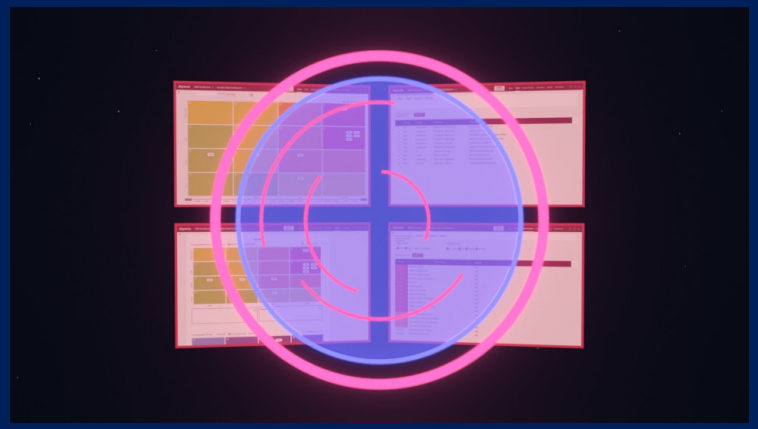
TEST, SHARE, USE

6. **Test in Preview**
   /idea A 6-second sting: "Hello DMS" among stars, a ring draws around it, a flash
7. **Check the answer**
   Shot Script draft, then one script: import motionkit … movie.run()
8. **Save and share**
   Save; Share with your developers (or each makes their own copy)
9. **Use it**
   Save the code in motionkit/examples/, then render with --still 3
10. **Keep it current**
    After kit changes: python gemini-gem/build_knowledge.py, re-upload

FILES YOU NEED

`erm-migration/motionkit/gemini-gem/GEM.md` text to copy `…/knowledge/motionkit-knowledge.md` file to upload

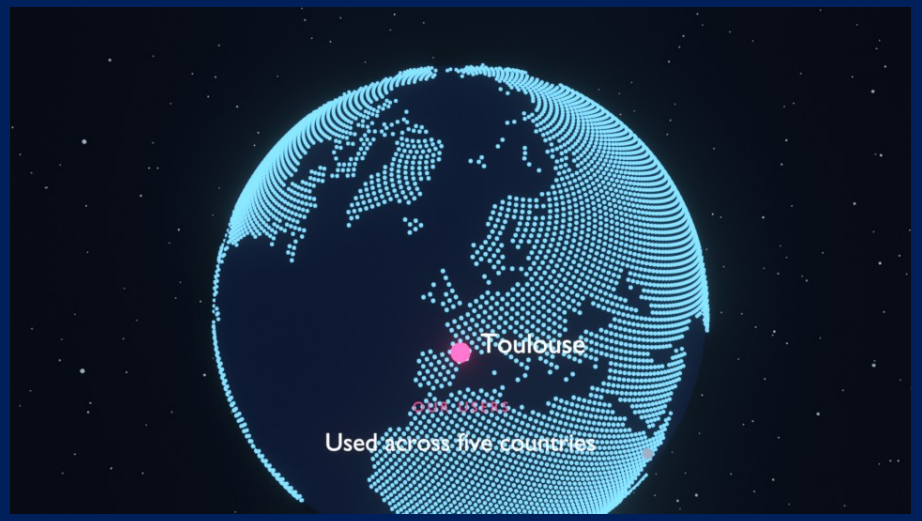
Three examples to copy from







02 · Product reveal (16 s) screen wall → portal → new app → chart and counter → end title. Shots, transitions, music.



03 · Globe tour (10 s)
The globe turns to each site; pins pop in with labels.

01 · Hello title (8 s)
The 25-line demo you just saw.

Tested on Blender 5, documented, easy to extend

5.0.1
Blender version it's tested on, with EEVEE

25
checks in the smoke test: every asset, motion, move, transition

11
cookbook recipes, each run by a test so the docs stay true

1
file holds every Blender-version difference (_compat.py)

DOCUMENTATION (docs/)

`getting-started.md`
install, run, first change

`concepts.md`
the six ideas behind the kit

`api-reference.md`
every class, method and argument

`cookbook.md`
copy-paste recipes

`shot-script.md`
the format for Gemini

`troubleshooting.md · extending.md`
fixes; adding your own assets

Why EEVEE

Our look is self-lit objects plus glow: no light bounces to simulate. EEVEE renders it in real time and looks the same as Cycles here, many times faster.

The kit switches off what we don't use (shadows, ray tracing, volumes), uses 64 samples for finals and 16 for previews, and does the glow in the GPU compositor.

Ten minutes, four steps

1 Render a frame

```
blender -b -P examples/01_hello_title.py -- --still
3.5
```

The picture appears in examples/render/

2 Change it

```
Text3D("Hello DMS", ...)  →  Text3D("Hello <your
team>", color="cyan")
```

Run the still again: the change shows in seconds

3 Ask the gem

```
/idea A 6-second sting: our logo appears among stars,
a ring draws around it, flash, end
```

Paste the code into examples/my_sting.py

4 Render a preview

```
blender -b -P examples/my_sting.py -- --preview --
render
```

Play the MP4

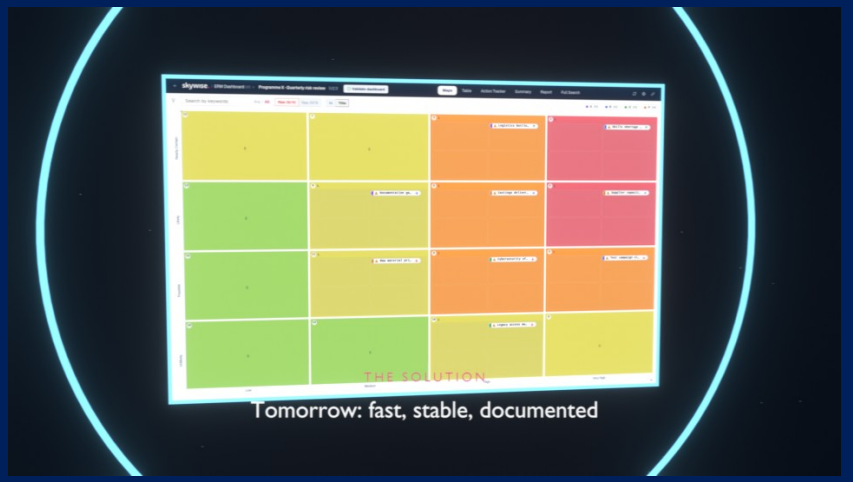
Your first video this week

1 **Install Blender 5**
blender.org, then check: blender --version

2 **Get the kit**
repo folder erm-migration/motionkit

3 **Run the demo**
`blender -b -P examples/01_hello_title.py -- --still 3.5`

4 **Open the gem**
`Gemini › Gems › motionkit Director (setup in gemini-gem/GEM.md)`



Questions?